

Supplementary Appendices: Critical Metrology

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A Detailed Derivation of the Cavity Electric Field and Hamiltonian

To bridge the gap between classical electrodynamics and the Quantum Rabi Model (QRM), we consider a 1D cavity of length L along the z -axis. The electromagnetic field must satisfy the wave equation and the boundary conditions established by perfectly conducting walls at $z = 0$ and $z = L$. This implies that the transverse electric field must vanish at these boundaries. We begin by defining a transverse vector potential $\vec{A}(z, t)$ that naturally satisfies these spatial boundary conditions:

$$\vec{A}(z, t) = \hat{x}A(t) \sin(kz), \quad (1)$$

where the wave number is $k = n\pi/L$ (with n being an integer). In the Coulomb gauge, the scalar potential is zero, meaning the electric and magnetic fields are derived entirely from the vector potential:

$$\vec{E}(z, t) = -\frac{\partial \vec{A}}{\partial t} = -\hat{x}\dot{A}(t) \sin(kz), \quad (2)$$

$$\vec{B}(z, t) = \nabla \times \vec{A} = \hat{y}kA(t) \cos(kz). \quad (3)$$

The total classical energy (the Hamiltonian) of the electromagnetic field is the volume integral of the energy density over the entire cavity volume V :

$$H_{Field} = \frac{1}{2} \int_V \left(\epsilon_0 |\vec{E}|^2 + \frac{1}{\mu_0} |\vec{B}|^2 \right) dV. \quad (4)$$

To evaluate this integral, we note that the spatial averages of the squared trigonometric functions over the cavity length are identical: $\int_0^L \sin^2(kz) dz = \int_0^L \cos^2(kz) dz = L/2$. Integrating over the cross-sectional area S (such that the total volume is $V = SL$), the energy simplifies to:

$$H_{Field} = \frac{1}{4} V \epsilon_0 \dot{A}^2(t) + \frac{1}{4\mu_0} V k^2 A^2(t). \quad (5)$$

Our goal is to map this electromagnetic energy to the Hamiltonian of a unit-mass quantum harmonic oscillator, which has the standard form $H = \frac{1}{2}(p^2 + \omega^2 q^2)$. To achieve this, we must define a new generalized coordinate $q(t)$ such that the kinetic energy term $\frac{1}{2}\dot{q}^2$ matches the electric energy term. We define:

$$\dot{q}(t) = -\sqrt{\frac{V\epsilon_0}{2}} \dot{A}(t) \implies q(t) = -\sqrt{\frac{V\epsilon_0}{2}} A(t). \quad (6)$$

Using the dispersion relation $\omega = kc$ and the speed of light $c = 1/\sqrt{\mu_0\epsilon_0}$, we can rewrite the magnetic energy term. The substitution confirms that the Hamiltonian perfectly matches the

harmonic oscillator form. More importantly, substituting $\dot{A}(t)$ back into our initial expression for the electric field yields the final required form:

$$\vec{E}(z, t) = \sqrt{\frac{2\omega^2}{V\epsilon_0}} q(t) \sin(kz) \hat{x}. \quad (7)$$

By promoting the classical variables $q(t)$ and $p(t)$ to quantum operators $\hat{q}$ and $\hat{p}$ satisfying $[\hat{q}, \hat{p}] = i$, and defining them in terms of bosonic creation and annihilation operators $\hat{q} = \sqrt{\frac{1}{2\omega}}(\hat{a} + \hat{a}^\dagger)$, we directly obtain the standard interaction term $H_{inter} = g(\hat{a} + \hat{a}^\dagger)\sigma_x$ utilized in the QRM.

B Positive Operator-Valued Measures (POVMs) vs. Projective Measurements

In the context of quantum metrology (Section 2.2.2), the ultimate goal is to extract the maximum amount of information about a target parameter θ encoded in a quantum state ρ_θ . In standard introductory quantum mechanics, measurements are typically described by Hermitian observables and their corresponding orthogonal projectors, known as von Neumann measurements. However, to find the absolute theoretical limit of precision—and to saturate the Quantum Cramér-Rao Bound (CRB)—we must employ a more generalized mathematical framework: Positive Operator-Valued Measures (POVMs). A POVM is defined as a set of Hermitian, positive operators $\{\Pi_x\}$ that sum to the identity:

$$\sum_x \Pi_x = \mathbb{I}, \quad \Pi_x \geq 0. \quad (8)$$

Unlike standard projectors, POVM elements are **not** required to be mutually orthogonal ($\Pi_x \Pi_y \neq \delta_{xy} \Pi_x$). This mathematical relaxation has profound physical implications. It allows for "fuzzy" measurements, or processes where the primary quantum system is coupled to an auxiliary quantum system (an ancilla) before a standard projective measurement is performed on the larger, combined Hilbert space. According to the generalized Born rule, the probability of obtaining a measurement outcome x is given by $p(x|\theta) = \text{Tr}(\rho_\theta \Pi_x)$.

The Quantum Fisher Information (QFI), $F_Q(\theta)$, is rigorously defined as the supremum of the Classical Fisher Information over *all possible* POVMs. By restricting ourselves solely to orthogonal projective measurements, we risk missing the optimal measurement strategy and failing to achieve the fundamental precision limit dictated by the QFI. Therefore, POVMs represent the most complete mathematical description of all physically permissible measurement strategies.

C Entanglement Entropy and the Von Neumann Formalism

In the Quantum Rabi Model, the ground state $|\psi\rangle$ exists within a composite, bipartite Hilbert space constructed from the tensor product of the cavity field mode and the two-level atom (qubit): $\mathcal{H}_{field} \otimes \mathcal{H}_{spin}$. To rigorously quantify the degree of quantum entanglement shared between the light (bosons) and the matter (spin), we utilize the Von Neumann Entropy, denoted as S_{VN} . First, we define the density matrix of the pure, global ground state as $\rho = |\psi\rangle\langle\psi|$. To evaluate the mixed state of the spin subsystem irrespective of the cavity field, we perform a partial trace over the infinite-dimensional bosonic degrees of freedom. This yields the reduced density matrix of the spin:

$$\rho_{spin} = \text{Tr}_{field}(|\psi\rangle\langle\psi|). \quad (9)$$

The Von Neumann Entropy is then calculated from this reduced density matrix using the formula:

$$S_{VN}(\rho_{spin}) = -\text{Tr}(\rho_{spin} \ln \rho_{spin}) = -\sum_i \lambda_i \ln \lambda_i, \quad (10)$$

where λ_i are the strictly positive eigenvalues of ρ_{spin} .

Physically, this entropy acts as an excellent probe for identifying phase transitions. In the normal phase, where the light-matter interaction is relatively weak, the system's ground state is nearly separable (e.g., $|0\rangle \otimes |\downarrow\rangle$), resulting in an entropy very close to zero. However, as the coupling strength increases and approaches the Quantum Critical Point (QCP) at γ_c , quantum fluctuations dominate. The entropy exhibits a sharp discontinuity and a prominent peak, indicating that the macroscopic cavity field and the single qubit have become highly entangled. This discontinuity, visualized in Section 6.4, serves as a definitive numerical signature of entering the superradiant phase.

D Critical Slowing Down: QRM vs. Thermodynamic Limit

A central caveat to the infinite susceptibility predicted by critical metrology is the phenomenon of Critical Slowing Down (CSD). CSD is a universal feature of continuous phase transitions wherein the characteristic timescale τ required for a system to recover thermodynamic equilibrium—or for dynamic perturbations to decay—diverges as it approaches the critical point. This dynamical divergence is fundamentally linked to the closing of the spectral energy gap Δ_{gap} between the ground state and the first excited state. Quantum mechanics dictates that the evolution timescale is inversely proportional to this gap: $\tau \propto 1/\Delta_{gap}$.

In standard many-body frameworks experiencing a Quantum Phase Transition (QPT), such as the prototypical Dicke Model (which features N two-level atoms interacting with a cavity mode), CSD manifests strictly in the Thermodynamic Limit (TDL). In the TDL, the number of constituents $N \rightarrow \infty$, and the spectral gap closes proportionally to $1/N$. Consequently, the time required for the system to settle into its critical ground state becomes strictly infinite only for an infinitely large system.

The Quantum Rabi Model represents a radical departure from this paradigm. It is a finite-sized system containing only a single atom ($N = 1$). In the QRM, criticality does not emerge from scaling particle numbers, but rather from the *Classical Oscillator Limit*, defined by the frequency ratio $\Delta/\omega \rightarrow \infty$. In this scenario, the manifestation of CSD occurs as the cavity frequency ω vanishes relative to the atomic transition frequency Δ . Here, the parameter scaling $\omega/\Delta \rightarrow 0$ effectively assumes the mathematical role of $1/N$ in thermodynamic systems.

For the field of critical quantum metrology, CSD presents a profound physical trade-off. While static theoretical models utilizing the Quantum Fisher Information predict arbitrarily high, diverging precision exactly at the QCP, the infinite relaxation time dictated by CSD implies that successfully preparing the system in that exact critical ground state would require an infinite amount of time. Thus, achieving enhanced sensing in any realistic experimental setup requires carefully navigating the balance between the metrological advantage of the critical divergence and the stringent temporal costs of state preparation.

E Numerical Convergence: Fidelity vs. Energy

When performing numerical diagonalization of the QRM, choosing a proper metric for convergence is critical. Initially, one might consider the convergence of the ground state energy. However, for metrological applications, this approach is often insufficient. The Quantum Fisher Information depends on the derivative of the state vector $|\psi\rangle$ and its internal structure. Near the

critical point, the energy can reach a stable value (converge) much faster than the wavefunction itself. If the Fock space truncation is too low, the energy may seem correct, but the state vector fails to capture the high-photon components necessary for accurate QFI calculation.

Therefore, we utilize the fidelity between consecutively truncated states:

$$\mathcal{F} = |\langle \psi_N | \psi_{N-1} \rangle|^2. \quad (11)$$

A fidelity close to unity ensures that adding more states to the basis does not significantly change the state vector, providing a much more stringent and reliable convergence criterion than the ground state energy.

F Physical Interpretation of the Unitary Displacement Transformation

In the superradiant phase of the QRM ($g > g_c$), the effective potential shifts from a single-well to a double-well structure. Physically, this means the ground state is no longer centered at the vacuum $|0\rangle$, but instead acquires a macroscopic photon occupation $|\alpha\rangle$. Attempting to describe this state using a standard Fock basis centered at the vacuum requires a prohibitively high truncation (hundreds or thousands of states) to capture the "tail" of the probability distribution.

The unitary displacement operator $D(\alpha)$ addresses this by shifting the origin of the phase space:

$$D^\dagger(\alpha)\hat{a}D(\alpha) = \hat{a} + \alpha. \quad (12)$$

By realigning the computational basis with the system's macroscopic mean field, we effectively describe small quantum fluctuations around a classical field. This concentrates the probability amplitude back into the lowest-lying Fock states of the transformed basis, allowing for high-fidelity numerical diagonalization with a drastically reduced Hilbert space dimension.

G Nature of Phase Transitions: Thermal vs. Quantum Fluctuations

To rigorously appreciate the nature of the superradiant phase transition in the Quantum Rabi Model (QRM), it is essential to distinguish between classical and quantum phase transitions based on their driving mechanisms.

Classical Phase Transitions (CPTs) occur at finite temperatures ($T > 0$) and are fundamentally driven by thermal fluctuations. The thermodynamic equilibrium state of a classical system is determined by the minimization of the Helmholtz free energy:

$$F = E - TS. \quad (13)$$

In this regime, the system experiences a competition between the internal energy E , which favors microscopic ordering (e.g., magnetic alignment in an Ising ferromagnet), and the entropy S , which favors disorder. As the temperature increases, the thermal energy $k_B T$ provided by the heat bath amplifies random microscopic fluctuations. At a critical temperature T_c , these thermal fluctuations overpower the ordering interaction energy, causing the macroscopic order parameter to vanish continuously, thereby triggering a second-order phase transition.

Conversely, Quantum Phase Transitions (QPTs) occur strictly at absolute zero ($T = 0$). At this limit, thermal fluctuations are entirely completely suppressed, and the entropic contribution to the free energy vanishes. The system is governed solely by the ground state of its quantum Hamiltonian $\hat{H}(g) = \hat{H}_0 + g\hat{H}_1$.

In a QPT, the transition is driven exclusively by *quantum fluctuations* arising from Heisenberg's uncertainty principle. These fluctuations are governed by non-commuting observables within the Hamiltonian ($[\hat{H}_0, \hat{H}_1] \neq 0$). As the non-thermal control parameter g (such as the light-matter coupling strength in the QRM) is varied, the relative dominance of these non-commuting terms shifts. At the Quantum Critical Point (QCP) g_c , the quantum fluctuations become strong enough to fundamentally reconstruct the ground state wavefunction, inducing a macroscopic change in the system's order parameter despite the complete absence of thermal kinetic energy. In the QRM, the classical scaling limit ($\omega/\Delta \rightarrow 0$) mathematically mimics the thermodynamic limit necessary for these zero-point fluctuations to induce a true non-analyticity in the ground state energy.

H Spontaneous $\mathbb{Z}_2$ Symmetry Breaking in the QRM

The emergence of the superradiant phase in the QRM is mathematically characterized by the spontaneous breaking of a discrete $\mathbb{Z}_2$ parity symmetry. The naked QRM Hamiltonian,

$$\hat{H} = \omega \hat{a}^\dagger \hat{a} + \Delta \hat{\sigma}_z + g(\hat{a}^\dagger + \hat{a}) \hat{\sigma}_x, \quad (14)$$

possesses a global parity symmetry associated with the total number of excitations in the system. We define the parity operator $\hat{\Pi}$ as:

$$\hat{\Pi} = e^{i\pi(\hat{a}^\dagger \hat{a} + \frac{\hat{\sigma}_z + 1}{2})}. \quad (15)$$

This operator applies a π phase shift for every photon in the cavity and flips the sign depending on the state of the two-level atom. It is straightforward to verify that the Hamiltonian commutes with the parity operator, $[\hat{H}, \hat{\Pi}] = 0$. Consequently, the total Hamiltonian is invariant under the $\mathbb{Z}_2$ transformation $\hat{a} \rightarrow -\hat{a}$ and $\hat{\sigma}_x \rightarrow -\hat{\sigma}_x$. Because of this commutation relation, the exact eigenstates of the system must possess a well-defined parity, either even ($\langle \hat{\Pi} \rangle = +1$) or odd ($\langle \hat{\Pi} \rangle = -1$).

Spontaneous Symmetry Breaking (SSB) occurs when the ground state of a many-body system (or an effective thermodynamic system, such as the QRM in the scaling limit) fails to exhibit the symmetries of its underlying Hamiltonian.

In the normal phase ($g < g_c$), the ground state is centered at the vacuum with zero macroscopic photon population ($\langle \hat{a} \rangle = 0$). This state respects the $\mathbb{Z}_2$ symmetry. However, in the superradiant phase ($g > g_c$), the effective potential develops a double-well structure. The ground state becomes doubly degenerate, with the two minima corresponding to macroscopic field amplitudes of opposite signs, $\langle \hat{a} \rangle = \pm\alpha$.

Because any physical measurement or minute perturbation will force the system to select one of these two degenerate minima, the realized ground state inevitably acquires a non-zero coherent field $\pm\alpha$. By settling into one specific well, the system "chooses" a sign, thereby spontaneously breaking the $\hat{a} \rightarrow -\hat{a}$ invariance. The order parameter of the transition (the photon number or the cavity displacement) emerges exactly as a consequence of this ruptured $\mathbb{Z}_2$ symmetry.

I Bias-Variance Decomposition of the Mean Squared Error

In the context of quantum parameter estimation and metrology, the performance and accuracy of an estimator $\hat{\theta}$ for a true underlying parameter θ is strictly quantified by its Mean Squared Error (MSE). The mathematical formulation of the MSE can be elegantly decomposed into two distinct components: the variance of the estimator and the square of its bias.

We begin with the fundamental definition of the MSE as the expected value of the squared difference between the estimator and the true parameter. By strategically adding and subtracting

the expected value of the estimator, $E_\theta[\hat{\theta}]$, within the quadratic term, the derivation proceeds as follows:

$$\begin{aligned}
\text{MSE}(\hat{\theta}) &= E_\theta \left[(\hat{\theta} - \theta)^2 \right] \\
&= E_\theta \left[\left(\hat{\theta} - E_\theta[\hat{\theta}] + E_\theta[\hat{\theta}] - \theta \right)^2 \right] \\
&= E_\theta \left[\left(\hat{\theta} - E_\theta[\hat{\theta}] \right)^2 + 2 \left(\hat{\theta} - E_\theta[\hat{\theta}] \right) \left(E_\theta[\hat{\theta}] - \theta \right) + \left(E_\theta[\hat{\theta}] - \theta \right)^2 \right] \\
&= E_\theta \left[\left(\hat{\theta} - E_\theta[\hat{\theta}] \right)^2 \right] + E_\theta \left[2 \left(\hat{\theta} - E_\theta[\hat{\theta}] \right) \left(E_\theta[\hat{\theta}] - \theta \right) \right] + E_\theta \left[\left(E_\theta[\hat{\theta}] - \theta \right)^2 \right]
\end{aligned}$$

Given that θ is a deterministic parameter and $E_\theta[\hat{\theta}]$ is a constant scalar value (not a random variable), the terms $\left(E_\theta[\hat{\theta}] - \theta \right)$ can be factored out of the expectation operators. Applying the linearity of the expectation value yields:

$$\begin{aligned}
\text{MSE}(\hat{\theta}) &= E_\theta \left[\left(\hat{\theta} - E_\theta[\hat{\theta}] \right)^2 \right] + 2 \left(E_\theta[\hat{\theta}] - \theta \right) E_\theta \left[\hat{\theta} - E_\theta[\hat{\theta}] \right] + \left(E_\theta[\hat{\theta}] - \theta \right)^2 \\
&= E_\theta \left[\left(\hat{\theta} - E_\theta[\hat{\theta}] \right)^2 \right] + 2 \left(E_\theta[\hat{\theta}] - \theta \right) \left(E_\theta[\hat{\theta}] - E_\theta[\hat{\theta}] \right) + \left(E_\theta[\hat{\theta}] - \theta \right)^2
\end{aligned}$$

The central cross-term clearly vanishes because $E_\theta[\hat{\theta}] - E_\theta[\hat{\theta}] = 0$. This leaves only the first and third terms:

$$\begin{aligned}
\text{MSE}(\hat{\theta}) &= E_\theta \left[\left(\hat{\theta} - E_\theta[\hat{\theta}] \right)^2 \right] + \left(E_\theta[\hat{\theta}] - \theta \right)^2 \\
&= \text{Var}_\theta(\hat{\theta}) + \text{Bias}_\theta(\hat{\theta}, \theta)^2
\end{aligned}$$

This rigorous decomposition demonstrates that the total estimation error arises from two sources: the intrinsic statistical dispersion of the measurement records (variance) and the systematic deviation of the estimator from the true value (bias). In the ultimate precision limits established by the Quantum Cramér-Rao Bound, it is customary to assume an unbiased estimator protocol ($\text{Bias}_\theta = 0$), allowing the MSE to reduce strictly to the fundamental variance of the quantum measurement.

J Cauchy-Schwarz Derivation of the Quantum Fisher Information Bound

In quantum estimation theory, the Quantum Fisher Information (QFI) establishes the ultimate precision limit for parameter estimation, bounding the Classical Fisher Information (CFI) maximized over all possible Positive Operator-Valued Measures (POVMs). This upper bound is rigorously derived using the Cauchy-Schwarz inequality over the space of matrices.

We begin with the general expression for the CFI associated with a quantum state ϱ_θ and a continuous set of POVM elements $\{\Pi_x\}$, expressed in terms of the Symmetric Logarithmic Derivative (SLD) operator L_θ :

$$F(\varrho_\theta) = \int dx \frac{\text{Re}(\text{Tr}[\varrho_\theta \Pi_x L_\theta])^2}{\text{Tr}[\varrho_\theta \Pi_x]}. \quad (16)$$

For any complex number z , the inequality $\text{Re}(z)^2 \leq |z|^2$ holds. Applying this to the numerator yields our first bound:

$$F(\varrho_\theta) \leq \int dx \frac{|\text{Tr}[\varrho_\theta \Pi_x L_\theta]|^2}{\text{Tr}[\varrho_\theta \Pi_x]}. \quad (17)$$

To bound the magnitude of the trace, we utilize the Hilbert-Schmidt inner product for matrices, defined as $\langle A, B \rangle = \text{Tr}[A^\dagger B]$. The Cauchy-Schwarz inequality for this inner product space dictates that $|\langle A, B \rangle|^2 \leq \langle A, A \rangle \langle B, B \rangle$.

Since both the density matrix ϱ_θ and the POVM elements Π_x are positive semi-definite operators, their square roots are well-defined and Hermitian. Utilizing the cyclic property of the trace, we strategically define two auxiliary operators to reconstruct the numerator:

$$A = \sqrt{\Pi_x} \sqrt{\varrho_\theta}, \quad (18)$$

$$B = \sqrt{\Pi_x} L_\theta \sqrt{\varrho_\theta}. \quad (19)$$

The inner product of these operators perfectly retrieves our target trace:

$$\langle A, B \rangle = \text{Tr}[A^\dagger B] = \text{Tr}[\sqrt{\varrho_\theta} \sqrt{\Pi_x} \sqrt{\Pi_x} L_\theta \sqrt{\varrho_\theta}] = \text{Tr}[\varrho_\theta \Pi_x L_\theta]. \quad (20)$$

Next, we evaluate the squared norms of A and B . For A , we obtain the denominator of our CFI expression:

$$\langle A, A \rangle = \text{Tr}[A^\dagger A] = \text{Tr}[\sqrt{\varrho_\theta} \Pi_x \sqrt{\varrho_\theta}] = \text{Tr}[\varrho_\theta \Pi_x]. \quad (21)$$

For B , exploiting the hermiticity of the SLD ($L_\theta^\dagger = L_\theta$) and the cyclic property of the trace, we find:

$$\langle B, B \rangle = \text{Tr}[B^\dagger B] = \text{Tr}[\sqrt{\varrho_\theta} L_\theta \Pi_x L_\theta \sqrt{\varrho_\theta}] = \text{Tr}[\Pi_x L_\theta \varrho_\theta L_\theta]. \quad (22)$$

Applying the Cauchy-Schwarz inequality, the squared trace is strictly bounded by the product of these norms:

$$|\text{Tr}[\varrho_\theta \Pi_x L_\theta]|^2 \leq \text{Tr}[\varrho_\theta \Pi_x] \text{Tr}[\Pi_x L_\theta \varrho_\theta L_\theta]. \quad (23)$$

Substituting this bound back into the modified CFI integral, the term $\text{Tr}[\varrho_\theta \Pi_x]$ gracefully cancels out with the denominator:

$$F(\varrho_\theta) \leq \int dx \frac{\text{Tr}[\varrho_\theta \Pi_x] \text{Tr}[\Pi_x L_\theta \varrho_\theta L_\theta]}{\text{Tr}[\varrho_\theta \Pi_x]} = \int dx \text{Tr}[\Pi_x L_\theta \varrho_\theta L_\theta]. \quad (24)$$

By the linearity of the trace, we commute the integral with the trace operation:

$$F(\varrho_\theta) \leq \text{Tr} \left[\left(\int dx \Pi_x \right) L_\theta \varrho_\theta L_\theta \right]. \quad (25)$$

Finally, we invoke the fundamental completeness relation of the POVM framework, which dictates that all measurement elements must integrate to the identity operator on the Hilbert space: $\int dx \Pi_x = \mathbb{I}$. Substituting the identity yields the ultimate bound:

$$F(\varrho_\theta) \leq \text{Tr}[\mathbb{I} \cdot L_\theta \varrho_\theta L_\theta] = \text{Tr}[\varrho_\theta L_\theta^2]. \quad (26)$$

This strict upper limit constitutes the formal definition of the Quantum Fisher Information, $F_Q[\varrho_\theta]$. Notably, it depends exclusively on the intrinsic properties of the quantum state and its parametrization, effectively removing any dependence on the specific measurement strategy chosen.